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GSR R-Notebooks: GSR Signal Quality

This article will walk you through the new GSR signal quality notebook and how you can use it to asses your GSR signal. This notebooks works GSR/EDA signals collected with Shimmer or Biopac.

1. Prerequisites

Supported stimulus types: All

Supported sensors: Shimmer, Biopac

Supported iMotions version: iMotions 8.1.4 and later

2. How to Analyze in iMotions Desktop

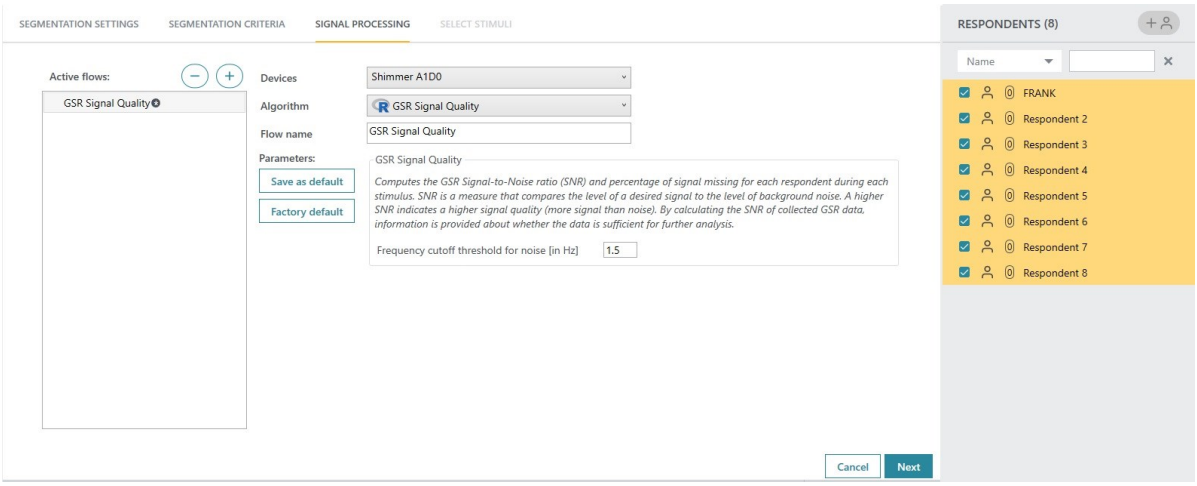
When you add your R-notebooks during your Signal Processing step, you can now add to analyze your GSR/EDA signal quality based on the R-notebook "GSR Singal Quality" which outputs two metrics:

(1) Signal-to-Noise ratio in decibels

(2) Signal missing in %

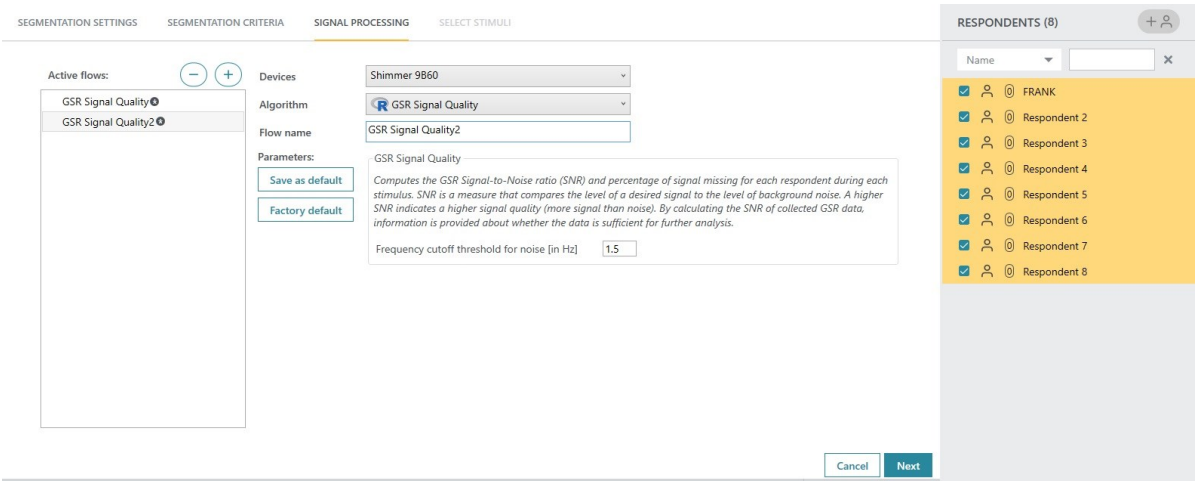
In order to access these metrics, please choose the GSR signal quality notebook during the Signal Processing step and add it to your active flows for the desired device. If you have multiple EDA/GSR devices, add the notebook separately for each device and give each active flow a unique name.

Single device:



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Multiple devices:



Continue with your analysis by clicking next and selecting the corresponding stimuli.

Once the algorithm has processed, you can go into the individual replays and review the Signal Quality for your respondents and stimuli. The signal quality is always calculated on the signal that is present, not considering signal lost. Hence always review these metrics together, to understand how much signal you have, and how much Noise this signal contains.

3. GSR Signal Quality Metrics and Algorithm

Metrics & Algorithm

Signal-to-Noise Ratio

Signal-to-noise ratio(abbreviatedSNRorS/N) is a measure that states how much signal relative to noise is contained in a signal or time series.SNR is defined as the ratio of signal power to the noise power, often expressed indecibels. A ratio higher than 1:1 (greater than 0dB) indicates more signal than noise.

The Signal Quality Notebook calculates the SNR based on the following steps as describedin [Schumm, J. 2008]:

- Retrieval of the calibrated GSR signal (expressed in microSiemens, µS) from the sensor.
- Determination of the sampling ratebased on timestamps
- Estimation of the Power Spectral Density (PSD) using the Fast Fourier Transform (FFT). Resulting frequencies are in range [0, Fn], where Fn (0.5 * samplingRate) is Nyquist Rate.

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- Based on the selected frequency cutoff parameter (here 1.5 Hz), separation between signal related PSDs (corresponding to frequencies inferior to 1.5 Hz) and noise related PSDs (corresponding to frequencies superior to 1.5 Hz). The default threshold value (5 Hz) has been chosen to preserve the electrodermal activity range of an adult human (0 to 1.5 Hz (Schumm et al. 2008)). This parameter can be decreased down to 1.5 Hz if wanted.
- Computation of the Signal-to-noise ratio (in decibel) using the following formula: SNR=10*log10(?signalPSD / ?noisePSD)
- Estimation of the proportion of signal missing based on the number of samples recorded, the duration of the stimulus and the estimated sampling rate of the device.

A negative value for the SNR indicates poor signal quality (more noise than signal). Also, based on our experience, a value superior to 20 dB seems sufficient to assess the good quality of the signal (signals with SNRs above 20 dB have less than 1% noise). However, it is hard to make a quality statement concerning values between 0 and 20. These stimulus/respondent pairs should be further investigated by having a look directly at the signal.

Signal Missing

Signal missing will give you the percentage of signal missing in relation to the stimulus length.

This is calculated based on the number of collected samples (and hence missing samples) over expected samples based on stimulus length expressed in %.

**This metrics used to be included in the GSR Peak Detection Notebook and Output. It is now part of the Signal Quality Output.

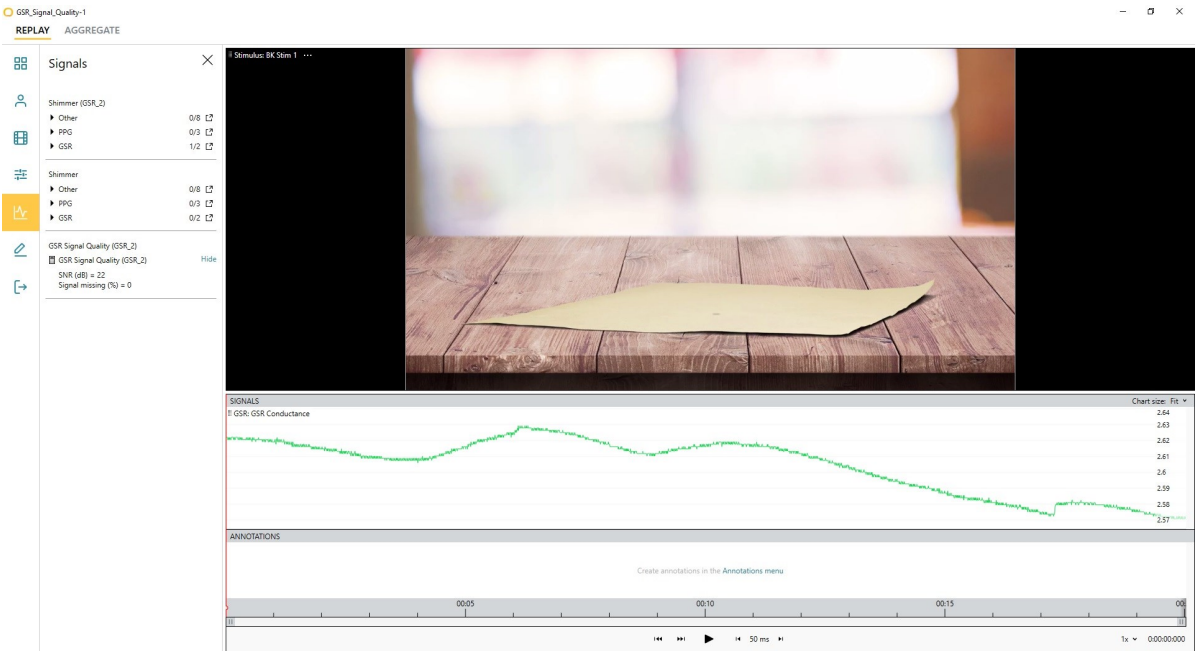
Examples

The following examples illustrate different scenarios to make it easier for combine these metrics in a sensible fashion.

Example 1: No Signal missing, SNR above 20 db

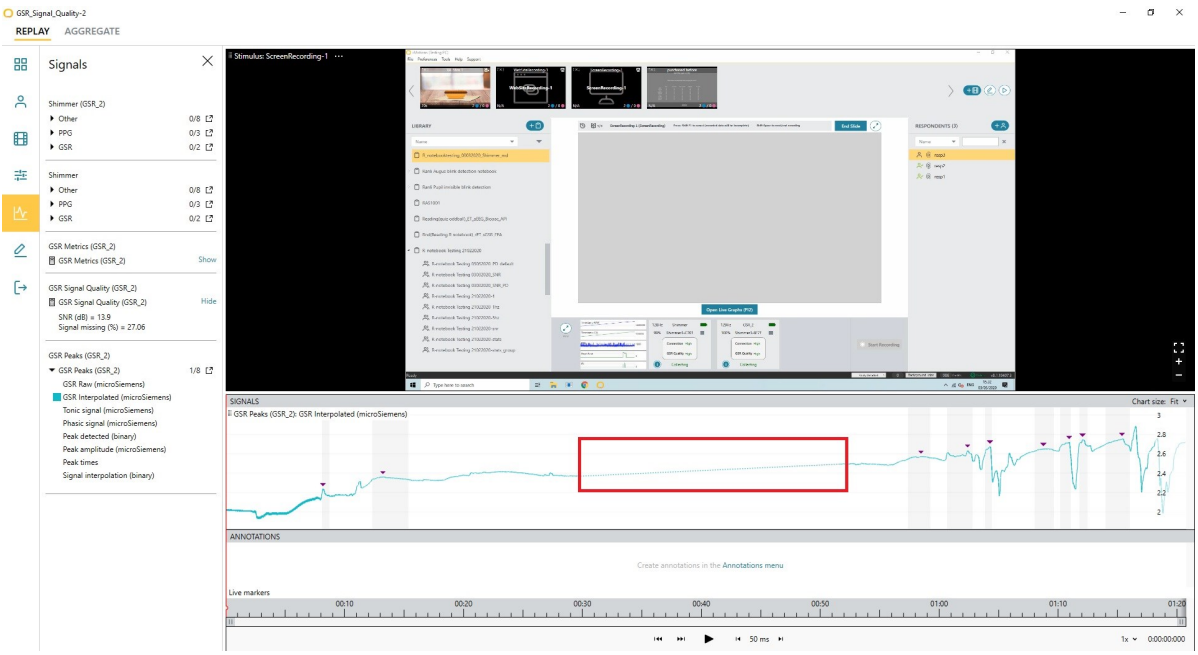
Here no signal has been lost and with a db above 20 you have less than 1% noise in the signal.

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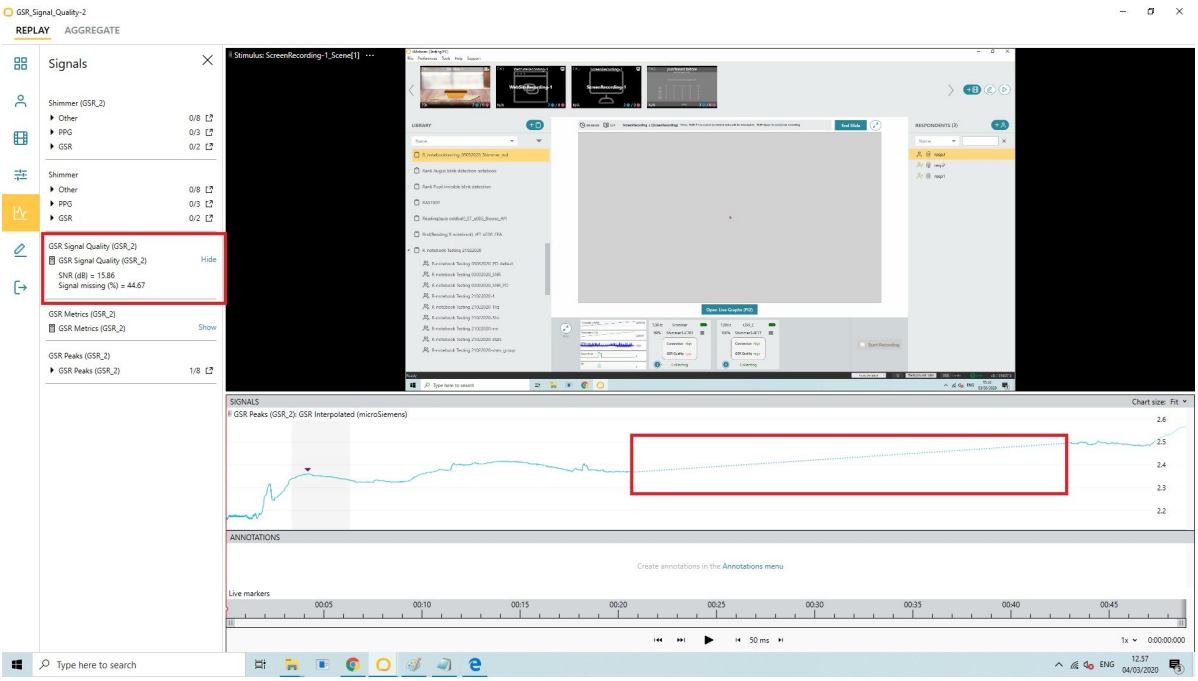
Example 2: Signal missing, SNR above 20 db

Here the SNR is good with db above 20 but parts of the signal are missing. This is also apparent in the peak detection where the missing signal is shown as a dashed line.



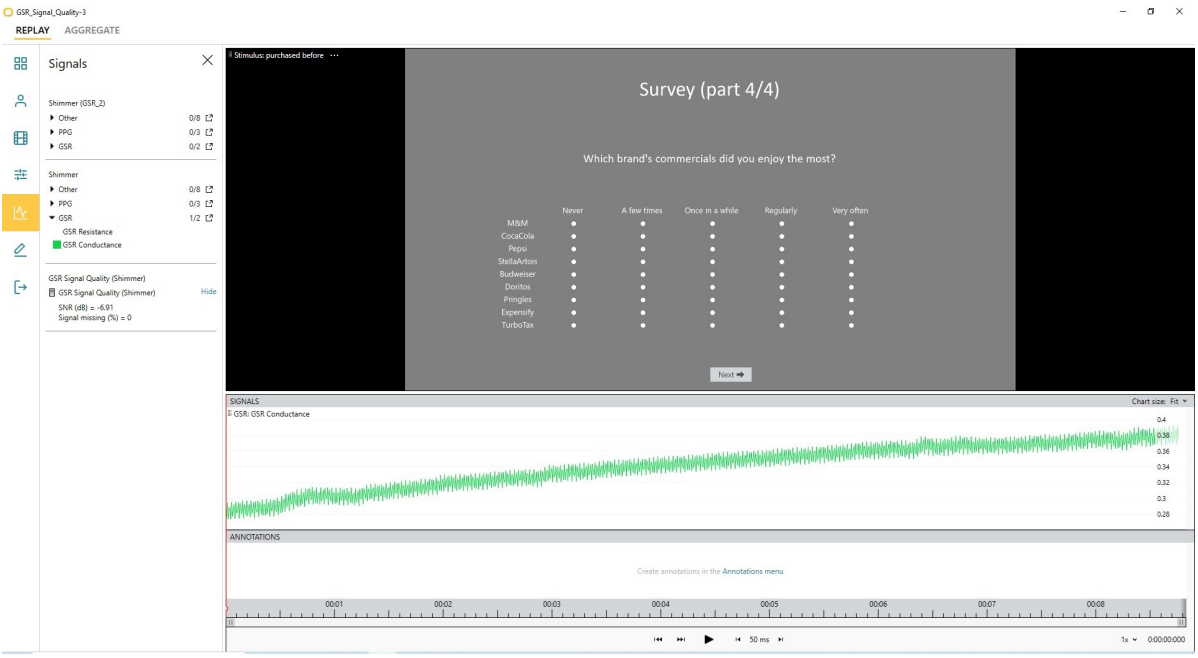
When you create a scene during the time window when parts of the signal is missing, the percentage of signal missing for that scene increase accordingly to the scene/stimulus length.

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Example 3: No Signal missing, SNR below 0 (more noise than signal)

Here no signal is missing but the measurement contains more noise than signal with -6.91db.



You can run the signal quality notebooks before running any other GSR notebook, or together with the peak detection and aggregation notebook.

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Depending on the signal quality you can exclude respondents from further analysis after exporting the GSR Signal quality and removing respondents accordingly. This can be done using the segmentation settings and a csv file which includes only respondents with a sufficient signal. For more information on how to do that, please review the following article:Respondent Segmentation using a CSV file.

4. The Outputs

In order to export the Signal quality including the SNR and Signal Missing, go to the Export dialogue and choose GSR Signal Quality Export under the R-processing tab. This will result in a txt file including all stimuli and respondent, SNR and signal missing.

Before you exclude respondents, make sure you have an overview of the average signal quality for each stimulus and respondent. If the overall quality is high, 20db is a valid threshold.

References

Schumm, J., M. Bachlin, C. Setz, B. Arnrich, D. Roggen, and G. Troster. 2008. “Effect of Movements on the Electrodermal Response After a Startle Event.” In *2008 Second International Conference on Pervasive Computing Technologies for Healthcare*, 315–18.

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